SVKM's NMIMS Deemed-to-be University
Mukesh Patel School of Technology Management and Engineering

					w.e.f 2025-26
Program: B Tech and MBA Tech All Programs				**Semester:** I/II	
Course: Programming with Python				**Code:**	
Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks - 50)**
1	2	0	2	Marks Scaled to 50	(Practical Examination) Marks Scaled to 50

Prerequisite: Nil

Course Objective
The course is designed to provide basic knowledge of Python programming and how to process, analyze data using python data structures.

Course Outcomes
After completion of the course, the student will be able to –
1. Implement programs using basic control and data structures (K3),
2. Implement programs using functions and files (K3),
3. Develop programs using arrays and DataFrame (K3)

Detailed Syllabus

Unit	Description	Duration
1	**Introduction to Python:** Python interpreter and its working, syntax and semantics, comments, indentation, variables, data types, operators and expressions, I/O operations, formatting print, generating random numbers, imports	02
2	**Program Flow Control:** Conditional blocks using if, else and elif, simple for loop, for loop using ranges, while loops, loop manipulation using pass, continue, break.	03
3	**Python Data Structures:** Strings, Lists, Dictionaries, Tuples, Sets, indexing and slicing, properties, operations and methods of these data structures, list and dictionary comprehension.	03
4	**Python Functions and Files:** Defining and calling functions, return, scope, function arguments (args and kwargs), recursive functions. File I/O – open, close, read, write and append operations.	03

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5	**Data handling using Numpy and Pandas:** Introduction Numpy Array, Difference between list and Numpy Array, Operations on 1-D and 2-D Numpy Array. Introduction to Pandas and DataFrame, Understanding Data, Importing and Exporting Data, Preprocessing of Data, Data Analysis.	04
	Total	**15**

Text Books:

1Dr. R. Nageswara Rao, *Core Python Programming*, 3rd Edition, Dreamtech Pres, Wiley Publication, 2021.
2Martin C. Brown, *Python: The Complete Reference*, 4th Edition, McGrawHill Education, 2018.

Reference Books:

1. Paul Barry, *Head first Python: A Brain Friendly guide*, 3nd Edition, O'Reilly publication, 2023.
2. Bill Lubanovic, *Introducing Python Modern computing in simple packages*, 3rd Edition, O'Reilly publication, 2019.
3. Dr. R. Nageswara Rao, *Core Python Programming*, 2nd Edition, Dreamtech Pres, Wiley Publication, 2018.

Laboratory/ Tutorial Work

10 experiments / Programming exercises (and a practicum where applicable) based on the syllabus

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